Math Inquirers Level 3: Scheme of Work

Holiday program for children whose school year covers Primary 3 topics • 10 lessons of 90 minutes • 2 lessons a week for 5 weeks • live online, up to 8 children

Approach. Enrichment, not extra drill: non-routine puzzles tagged to the school syllabus. Every lesson follows the same routine: warm-up (10 min), explore with objects or drawings (15), mini-lesson (10), brain break (5), worksheet in pairs then alone (30), talk time (10), exit ticket and journal stamp (10).
Habits. Notice it. Test it. Explain why.

Lesson	Title and syllabus topic	Objective	Skills and heuristics practised	Exit ticket
1 Week 1A	**Number Sleuths** Numbers to 10 000	Read and build 4-digit numbers by place value; make the greatest, smallest, odd, even and closest numbers from digit cards; list every number in order; solve clue puzzles and continue number patterns.	Place value to 10 000, renaming (15 hundreds = 1 thousand 5 hundreds), making a list in order, looking for a pattern, using clues one at a time, explaining "I have them all"	Use the cards 6, 0, 3, 8 once each. Write the smallest 4-digit number and the greatest odd 4-digit number.
2 Week 1B	**Cryptarithm Crackers** Addition and Subtraction	Find missing digits in 4-digit addition and subtraction by working column by column from the ones, check subtraction by adding, work backwards in magic squares and number pyramids, and use guess, check and improve for digit-card and letter puzzles.	Column addition and subtraction with regrouping, using the opposite operation (work backwards), guess and check then improve, reasoning about carrying, explaining "this is the only answer"	Find the missing digit A: 5A3 + 126 = 689
3 Week 2A	**Times Table Treasure** Tables of 6, 7, 8 and 9	Find and explain patterns in the 6, 7, 8 and 9 times tables, use last-digit patterns, split an array into facts we know, and solve grid and list puzzles in a careful order.	Look for a pattern, make a systematic list, use known facts, explain why	What is the last digit of 6 × 47? Explain how you know without working out 6 × 47.
4 Week 2B	**Model Makers** Word Problems: Bar Models	Draw bar models for part-whole, comparison, times-as-many and before-and-after word problems, and use them to solve two-step problems and check the answers.	Draw a model, before and after, look for what stays the same, check the answer	Ali has 3 times as many stickers as Ben. They have 80 stickers altogether. How many stickers does Ali have? Draw the model.
5 Week 3A	**Remainder Riddles** Division	Divide with remainders, check that the remainder is smaller than the number we divide by, decide what the leftover means in a story, work backwards from groups and leftovers, and solve "what number am I?" riddles by making a list.	Division with remainders, interpreting the remainder (one more group or leave it out), working backwards (groups × size + leftover), making a systematic list and testing each clue, making a table	29 children go on a ride. Each car holds 4 children. How many cars are needed? Write the division and explain what the leftover means.
6 Week 3B	**Fraction Walls** Fractions	Use a fraction wall to find equivalent fractions, compare and order fractions by making the parts the same size, and add and subtract like and related fractions within one whole.	Drawing a model (fraction wall, bar), equivalent fractions (multiply top and bottom by the same number), comparing with the same bottom number or the same top number, adding related fractions, finding all fractions between two fractions	Ali says 1/4 + 1/8 = 2/12. Is he right? Write the correct answer and show how you know.
7 Week 4A	**Area Architects** Area and Perimeter	Find the area and perimeter of figures made of squares and rectangles, find every rectangle with a given perimeter by listing in order, and use “change the shape” (push the steps out; cut up or fill in) to find perimeters and areas quickly.	Systematic listing, changing the shape, spatial reasoning, explaining	A rectangle has a perimeter of 12 cm. Its length is 4 cm. What is its area?

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8 Week 4B	**Angle and Line Detectives** Angles, Perpendicular and Parallel Lines	Compare angles with a right angle using a folded-paper checker, count the angles at a point and the right angles in shapes and figures in an organised way, find a pattern in the counts, and find and draw perpendicular and parallel lines.	Organised counting, pattern spotting, checking with a tool, explaining	Which of these letters have BOTH a pair of parallel lines and a pair of perpendicular lines: F, V, H, Z?
9 Week 5A	**Measure and Money Masters** Length, mass, volume, money and time	Change compound units (m and cm, kg and g, ℓ and ml, $ and ¢) to one unit to compare, add and subtract; plan a budget; find durations and start and end times by drawing a time line, in 12-hour and 24-hour time.	Converting units, comparing, budgeting, drawing a time line, working backwards, explaining	A film starts at 3:30 p.m. and lasts 1 h 20 min. When does it end?
10 Week 5B	**Challenge Day** Mixed puzzles: bar graphs and logic	Choose a way to start a puzzle (draw it, list it, make a table), solve a logic grid with ticks and crosses, read and complete a bar graph with a scale, and explain the thinking.	Choosing a strategy, logical reasoning, reading scales, listing systematically, perseverance, explaining	Ali is 4th from the front and 4th from the back of a queue. How many people are in the queue? Which start did you use?

Assessment and progress

Each lesson has a 2-page worksheet (Start here), Think harder and Go further pages, an exit ticket and an Inquiry Journal stamp. Tutors record S (secure), D (developing) or H (needs help) for each child on the progress tracker in the tutor guide. Parents receive a short note each week and the Weekly Dose of Thinking Math practice pack.

Materials

Printed student workbook with cut-outs, pencil, scissors and paper; a laptop or tablet with a camera. Tutors use the slides for each lesson and the tutor guide scripts.